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A hybrid optimization-simulation approach for robust weekly aircraft routing and retiming

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ABSTRACT

We address the robust weekly aircraft routing and retiming problem, which requires determining weekly schedules for a heterogeneous fleet that maximizes the aircraft on-time performance, minimizes the total delay, and minimizes the number of delayed passengers. The fleet is required to serve a set of flights having known departure time windows while satisfying maintenance constraints. All flights are subject to random delays that may propagate through the network. We propose to solve this problem using a hybrid optimization-simulation approach based on a novel mixed-integer nonlinear programming model for the robust weekly aircraft maintenance routing problem. For this model, we provide an equivalent mixed-integer linear programming formulation that can be solved using a commercial solver. Furthermore, we describe a Monte-Carlo-based procedure for sequentially adjusting the flight departure times. We perform an extensive computational study using instances obtained from a major international airline, having up to 3387 flights and 164 aircraft, which demonstrates the efficacy of the proposed approach. Using the simulation software *SimAir* to assess the robustness of the solutions produced by our approach in comparison with that for the original solutions implemented by the airline, we found that on-time performance was improved by 9.8–16.0%, cumulative delay was reduced by 25.4–33.1%, and the number of delayed passengers was reduced by 8.2–51.6%.
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1. Introduction

Over the last few decades, the airline industry has become an essential pillar of the world's economy, as demonstrated by the ever-increasing number of air passengers and the volume of air freight. As a case in point, data disclosed by the International Air Transport Association (IATA) reveals that airlines transported nearly 3.5 billion passengers in 2015, and carried some 50 million metric tons of cargo across an impressive network of 50,000 routes. This huge connectivity has an overwhelming impact on the global economy, as demonstrated by the \$2.4 trillion of business activity shared by nearly 260 airlines.[1] These companies are continuously striving to enhance their profitability in a highly competitive environment by providing both high-quality and low-cost services, where legacy airlines have played a pioneering role in extensively employing

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[1] http://www.iata.org/about/Documents/iata-annual-review-2015.pdf.

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Operations Research (OR) to efficiently manage their processes. In particular, OR analysts have spent a great deal of effort in developing effective models and algorithms for optimizing the following major airline planning processes:

1. Schedule design (Jiang and Barnhart, 2009; Sherali et al., 2011): This process is concerned with building a profitable flight timetable to serve the targeted markets. It requires specifying flight origins and destinations together with the corresponding frequencies and departure times, while taking into consideration market profitability, resource availability, and regulations. The generated schedule constitutes the basis for all subsequent airline operations.
2. Fleet assignment (Barnhart et al., 2003; Sherali et al., 2006; Barnhart et al., 2009; Dožić and Kalić, 2015): This deals with matching the expected number of passengers to the correct seat capacity, while seeking to minimize passenger spill as well.
3. Aircraft routing (Clarke et al., 1997; Gopalan and Talluri, 1998; Lacasse-Guay et al., 2010): This planning process aims to determine for each individual aircraft the most profitable routing while satisfying maintenance restrictions. In this context, an aircraft routing refers to a time-compatible sequence of flight legs, where two consecutive flight legs are connected by their departure and arrival stations.
4. Crew scheduling (Desaulniers et al., 1998; Cohn and Barnhart, 2003): This process requires assigning qualified cockpit and flight attendant crews to each scheduled flight leg while satisfying a wide range of mandatory duty rules.

In this context, a seemingly paradoxical fact is that, despite their overwhelming effectiveness, optimized schedules are scarcely implemented as planned. The predominant reason is that airlines operate in an extremely dynamic and unstable environment while dealing with many uncertain parameters and factors outside their control. Consequently, unexpected incidents often lead to disrupting the planned schedules, thereby forcing airlines to rely on contingency plans to implement changes to the publicized schedule. The causes of disruptions are numerous and include aircraft breakdowns, air-traffic congestion, crew shortages, aircraft arriving late, and inclement weather conditions, to quote just the most common ones. It is well documented that these disruptions have been plaguing airlines worldwide, and that the number of disrupted flights is ever on the rise due to the dramatic growth in the airline industry. In 2014, it has been reported that 23.75% of US airlines' flights were delayed by more than 15 min.[2] Consequently, flight delays place a significant strain on the air transportation sector. Ball et al. (2010) reported that the cost of domestic flight delays put a $32.9 billion dent into the U.S. economy in 2007.

In this state of art, the airline industry is in greater need than ever to implement robust optimization approaches and paradigms to mitigate, in a cost-effective manner, the lingering negative impact of disruptions. Thus far, two broad classes of strategies have been investigated in the literature: proactive strategies and reactive strategies. On one hand, the class of proactive strategies refers to a set of approaches that aim at building robust schedules that are less vulnerable to disruptions or are easy to recover. In other words, a proactive strategy seeks to attribute, at the planning stage and thereby in anticipation of any disruption, some robustness features into the planned airline schedules. On the other hand, the class of reactive strategies refers to a set of recovery policies that seek, by using optimization methodologies, to repair the delayed schedules after the occurrence of a disruption in a quick and cost-effective way. So far, a wide range of reactive strategies have been adopted by airlines including aircraft rerouting, flight cancellation, crew rescheduling, and transporting disrupted passengers to their final destinations using alternative routes and/or airlines. In-depth surveys of reactive strategies can be found in Ball et al. (2007) and Barnhart (2009).

In this paper, we address the following tactical aircraft routing problem that is usually solved well in advance of departures. Given a one-week set of flights to be flown during a specific season of the year, together with (narrow) time-windows associated with the different departure times, the problem requires setting a departure time for each flight and deriving periodic maintenance feasible aircraft routes while covering all flights. The objective is to build robust aircraft schedules that are less sensitive to delays caused by late-arriving aircraft. Our motivation for emphasizing this latter source of delays stems from the fact that this cause is predominantly responsible for delays worldwide. Indeed, 39.8% of the total delay recorded by US carriers in 2015 were caused by late-arriving aircraft.[3] More specifically, we describe a proactive approach for enhancing the robustness of a weekly flight schedule by judiciously reallocating aircraft connection buffer times while accommodating maintenance constraints. In this effort, we make the following contributions:

1. We propose a compact mixed-integer nonlinear programming (MINP) model to solve the robust weekly aircraft maintenance routing problem. To the best of our knowledge, this is the first compact model for this weekly variant, and is derived by extending the daily model of Haouari et al. (2013).
2. We apply the Reformulation-Linearization Technique (RLT) of Sherali and Adams (1990, 1994) to provide an equivalent mixed-integer linear programming formulation than can be solved using a commercial solver.
3. We describe a hybrid optimization-simulation approach that delivers robust aircraft routes having enhanced on-time performance, while accommodating maintenance constraints and stochastic delays. The proposed approach combines mixed-integer programming with Monte-Carlo simulation to derive robust aircraft schedules.

[2] http://www.bts.gov/xml/ontimesummarystatistics/src/index.xml.

[3] Understanding the Reporting of Causes of Flight Delays and Cancellations, {http://www.bts.gov/help/aviation/html/understanding.html}.

4. We perform an extensive computational study using instances based on data provided by a major international airline, and having up to 3387 flights and 164 aircraft. The quality of the proposed solutions is assessed using *SimAir* as a simulation tool (Rosenberger et al., 2002).

The remainder of this paper is organized as follows. In Section 3, we provide a review of the literature pertaining to both airline maintenance planning and airline robust planning models. In Section 4, we describe the nonlinear compact weekly aircraft routing model, together with a lifted linearized version. Next, in Section 5, we describe a simulation-based flight retiming procedure. In Section 6, we present the results of our computational study. Finally, we provide a summary of our contributions and outline some future research directions.

2. Literature review

In this section, we successively outline the literature pertaining to robust aircraft routing and aircraft maintenance routing problem.

2.1. Robust airline planning

Aircraft routing robustness may be achieved through enhancing its *flexibility*or its *stability*. In this context, a schedule is considered as flexible if it exhibits an increased number of opportunities to easily recover from a disruption. So far, the most popular approaches used to generate flexible plans include, but not restrictively: (i) Aircraft swap opportunities (Ball et al., 2007; Burke et al., 2010; Ionescu and Kliewer, 2011), (ii) Insertion of extra flights (Tekiner et al., 2009; Muter et al., 2013), and (iii) Short cycles and hub connectivity (Rosenberger et al., 2004). On the other hand, a schedule is considered as stable if it is likely to remain feasible and cost-effective in case of (minor) disruptions. A common practice to build stable airline schedules consists in inserting slacks (or buffer times) into the most vulnerable connections. In this vein, Lan et al. (2006) proposed two distinct models for building stable aircraft routes. In the first model, they focused on minimizing the total expected propagated delay of the developed aircraft routes. The second model selects flight departure times to minimize the expected number of disrupted passengers while maintaining the current fleeting, aircraft routes, and passenger itineraries. Moreover, Yen and Birge (2006) proposed a stochastic integer programming model to solve the airline crew scheduling problem. They developed an algorithm that identifies flight connections that yield highly costly delays, and determines slacks to be inserted. Alternatively, AhmadBeygi et al. (2010) proposed two flight retiming models to redistribute slacks between flight connections that are most prone to delay propagation. In so doing, they reduced delay propagation while preserving crews and fleet restrictions, thereby leaving revenue projections unchanged. Subsequently, Dunbar et al. (2012) investigated the integrated robust aircraft routing and crew scheduling problem in order to capture delays that are transferred between late running aircraft and crews. They introduced an iterative approach to accurately estimate propagated delays and to produce robust plans for aircraft and crews. Dunbar et al. (2014) extended the former approach by considering stochastic delay information. Yan and Kung (2016) introduced a robust optimization-based formulation for the aircraft routing problem that explicitly considers correlation in flight leg delays. The authors employed a column-and-row generation framework to solve the considered problem. Aloulou et al. (2013) addressed the issue of constructing robust airline schedules through flight retiming and aircraft rerouting techniques. The authors designed a mixed-integer programming (MIP) model that judiciously allocates slack times across connecting fights. The proposed objective function maximizes a surrogate robustness measure that quantifies the ability of a schedule to absorb delays. Furthermore, Ben Ahmed et al. (2016) addressed the same problem and developed a two-level solution strategy that embeds a simulation-optimization procedure within an evolutionary algorithm. The simulation optimization procedure invokes a quadratic programming approach and a Monte-Carlo simulation procedure, together with a Particle Swarm Optimization algorithm (PSO) to compute the buffer times to be allocated to the frequently delayed connections. Alternatively, Cadarso and Marín (2013) described an integrated robust approach for solving the schedule design and fleet assignment problem. They sought to reduce the expected number of misconnected passengers by adding extra buffers to the most likely missed connections for passengers. Recently, Jamili (2016) designed a Simulated Annealing (SA) based heuristic to solve the integrated aircraft routing and scheduling problem with fleet assignment considerations. The output solution consists of aircraft routes together with flight departure times that incorporate sufficient buffer times between each pair of connecting flights.

2.2. Aircraft maintenance routing

The aircraft maintenance routing problem (AMRP) consists in building a sequence of flights to be flown consecutively, and that are served by a fleet of aircraft in which each aircraft must undergo a maintenance check within a preset number of days (or, accumulated flying hours). The objective is to minimize maintenance costs subject to specific operational constraints (e.g., availability of maintenance stations, availability of maintenance slots, etc.). There are two different variants (namely, tactical and operational) of the aircraft maintenance routing problem have been addressed in the literature. The tactical problem is meant to be solved several months before the departure dates, and aims at producing cyclic (steady state) schedules. It requires that each aircraft will serve the same sequence of flights, regardless of its initial location and initial accumulated flying time. In contrast, the operational problem is not periodic and explicitly considers the aforementioned aircraft operational restrictions.

2.2.1. Tactical models

The tactical aircraft maintenance routing problem was first investigated by Kabbani and Patty (1992) who developed a set-partitioning model to construct maintenance feasible routes that permit aircraft to ground overnight at a compatible maintenance station. Talluri (1998a) tackled the four-day maintenance routing problem while satisfying aircraft balance check requirements. The authors relied on a polynomial-time algorithm to build line-of-flights (LOF) that satisfy maintenance requirements. Mak and Boland (2000) formulated the AMRP as an asymmetric traveling salesman problem with replenishment arcs, and used an oriented simulated annealing solution strategy to obtain feasible solutions. Afsar et al. (2006) investigated the 7-day aircraft maintenance routing problem that maximizes aircraft utilization. Since this problem is typically very difficult to solve, the authors developed a longest path based heuristic approach. Liang et al. (2011) proposed a new mixed-integer linear programming formulation to solve the daily aircraft maintenance routing problem. The model was evaluated using four instances, and was found to be comparable to other existing formulations in the literature. Later, Liang and Chaovalitwongse (2012) extended this formulation to the weekly aircraft maintenance routing problem (WAMRP) via a novel rotation tour network representation. Moreover, Haouari et al. (2013) designed a compact node-arc flow formulation to solve the daily aircraft maintenance routing problem. The proposed model inherits certain nonlinear relationships that were linearized and reformulated using an RLT approach. Recently, Gavranis and Kozanidis (2015) addressed the flight and maintenance planning (FMP) problem through a mixed-integer linear programing model. The objective was to maximize the fleet availability of a unit of aircraft.

2.2.2. Operational models

The operational aircraft maintenance routing problem (OAMRP) has recently gained an increased interest, since long-term (tactical) plans are difficult to implement due to irregular events (aircraft diversions, early maintenance checks, etc.). Thus, it has became imperative to explicitly consider aircraft operational restrictions within the built-ahead plans. Sarac et al. (2006) addressed the operational aircraft maintenance routing problem on a daily basis, where the objective was to minimize the unused legal flying hours (defined as the maximum flying time between consecutive maintenance checks minus the actual accumulated flying time since the last maintenance check). Moreover, Orhan et al. (2012) designed an integer linear goal programming approach to solve the OAMRP, while maximizing aircraft utilization rates. Similarly, Başdere and Bilge (2014) addressed the operational aircraft maintenance routing problem. They proposed a multi-commodity flow model that aims to minimize the total unutilized legal flying time of a critical aircraft, which is one that might undergo maintenance during the one-week planning horizon. The authors provided two variants of the problem: an uncapacitated version that was solved using a commercial solver and a capacitated one that was solved using a simulation annealing heuristic. Recently, Al-Thani et al. (2016) investigated the OAMRP in order to minimize the total legal remaining flying time. They presented a mixed integer programming model that includes a polynomial number of variables and constraints. In addition, the model was embedded with a very large-scale neighborhood search (VLNS) algorithm to effectively solve large-scale data instances. In contrast to Başdere and Bilge (2014), Al-Thani et al. (2016) did not set any restrictions on the number of maintenance checks that an aircraft can undergo during the planning horizon.

2.2.3. Integrated models

To achieve higher-quality schedules, the AMRP has also been integrated with one or more additional planning stages. In this regard, the integrated fleet assignment and aircraft maintenance routing problem was investigated by Clarke et al. (1996), Barnhart et al. (1998) and Liang and Chaovalitwongse (2012). Furthermore, the integrated crew pairing and aircraft maintenance routing problem was addressed by Cordeau et al. (2001), Cohn and Barnhart (2003), Mercier et al. (2005) and Díaz-Ramírez et al. (2014). Recently, Shao et al. (2017) proposed an integrated model along with associated solution methodologies to solve fleet assignment, aircraft routing and crew pairing problems within a unified framework.

2.2.4. Robust aircraft maintenance routing

To the best of our knowledge, the only work pertaining to robust aircraft maintenance routing is due to Liang et al. (2015). The authors, proposed a novel line-of-flights (LOF) network model to solve the robust weekly aircraft maintenance routing problem using a two-stage column generation framework that selects line-of-flights exhibiting optimal buffer time arrangements.

3. Problem description and overview of the solution strategy

3.1. Problem description

Due to significant seasonality in demand, most airlines implement seasonal schedules that differ in terms of planned routes, flight frequencies, departures times, and fleeting decisions. Each of these seasonal schedules consists of a typical weekly schedule that is planned to be operated during several consecutive weeks. In this problem, we investigate a *tactical* aircraft routing problem that is essential to the process of designing a plan for a given season. This problem is intended to be solved after the fleet assignment, but well ahead of implementing operations. We assume that we are given a set L of n flight legs that are scheduled to be flown during a one-week planning horizon (that is, 10,080 min). Typically, the flight schedule includes thousands of domestic and international flight legs, and is assumed to repeat itself every week. Each flight leg $j \in L$ is characterized by the following attributes: (i) a departure station DS_j, (ii) an arrival station AS_j, (iii) a departure time $DT_j \in [0, 10{,}080)$, and (iv) an arrival time $AT_j \in [0, 10{,}080)$. Note that the day of operation of flight j is defined as $DO_j \equiv \left\lceil \frac{DT_j}{1440} \right\rceil$. We assume that, at this stage, the departure and arrival times are still provisional and might be subject to *minor* adjustments. That is, the departure time of each flight can be suitably shifted backward or forward within a predefined narrow time gap, typically ranging between 10 and 20 min. As we shall see later, this additional flexibility effectively promotes the generation of robust aircraft routes. At this point, it worth emphasizing that while significant alterations to the flight departure times could affect the demand (and thereby the revenue) for those flights, it is realistic to assume (as we do) that minor changes will not affect the demand.

The set of flights is partitioned into a family of C subsets (or *flight clusters*) where each cluster corresponds to a set of similar flights that originate from and arrive at the same stations and are scheduled at the same time for the different weekdays. Hence, depending on the frequency of each flight, each cluster may include from one to seven flights having the same identifier. For instance, a cluster may include all the three flights of Flight QR6368 connecting Doha to Milan that is regularly scheduled on Monday, Wednesday, and Saturday, departing at 9:10 am. The flight legs are served by a heterogeneous aircraft fleet that includes K aircraft types. Each aircraft type k includes NA_k aircraft, and is characterized by a specific seating configuration, a turn-time θ_k that corresponds to the minimum idle time that must elapse between a landing and the next takeoff, and a maintenance time M_k. This time corresponds to the duration of a mandatory maintenance check that must be periodically achieved, at a suitable maintenance station, before accumulating a specific number Ω_k of flying days (typically, Ω_k ranges from 4 to 7 days). We assume that the fleeting decisions were made beforehand at a previous stage, and are thereby part of the problem input. Hence, each flight leg should be served by a specific aircraft type. Note that flights that belong to the same flight cluster are not necessarily assigned to the same aircraft type. For instance, if the demand on some daily flight experiences a surge on each weekend then a narrow-body aircraft might be assigned during the weekdays, and a wide-body during the weekends. Furthermore, we are given a set S of stations where maintenance can be performed.

The *Robust Weekly Aircraft Routing and Retiming Problem* (RWARR), investigated in the present paper, requires: (*i*) setting the flight departure time of each flight, and (*ii*) building a route for each aircraft, while accommodating the following constraints:

- All flights that belong to the same flight cluster should depart at the same time (modulo 24 h) within the specified time windows. This synchronization constraint is mainly motivated by marketing purposes since airlines typically consider that scheduling similar flights at the same departure time positively impinges on their profitability. In the airline operations research literature, synchronization constraints were incorporated by several authors within fleet assignment and aircraft routing models. See for example Desaulniers et al. (1997), Ioachim et al. (1999), Bélanger et al. (2006) and Zeghal et al. (2010).
- Each flight should be covered by exactly one aircraft having a type that is ascribed by the fleet assignment decisions.
- For each aircraft type, the number of aircraft that are operated is fixed.
- Each aircraft should undergo a maintenance check, and spend the required maintenance duration, before accumulating a specific number of flying days.
- The routing should be periodic and thereby repeats itself every week.

Assume that $S = (\boldsymbol{\tau}, \mathbf{x})$ is the incidence vector of a feasible solution, where $\boldsymbol{\tau}$ and $\mathbf{x}$ represent the incidence vectors of the departure times and the aircraft routes, respectively. For each week of operation, the number of on-time flights (i.e., flights departing at the planned time) depends both on the solution S and a large number of uncontrollable events that may be caused by numerous random sources. Some of these sources are independent (i.e, *primary delays* caused by aircraft breakdown, station facilities congestion, crew sickness, or adverse weather), while the others are dependent (*reactionary delays*, caused by aircraft or crew arriving late, which propagate through the flight network). Define $PM(S)$ as the realized solution's on-time performance which is computed as the percentage of the number of on-time flights, over a one-week period of time, with respect to the total number of scheduled flights (though, in practice, a threshold delay value of 15 min may be set for considering a flight to be late). The on-time performance is a measure of the ability of the airline to deliver timely services and is often regarded as an airline performance measure of paramount importance. Hence, the objective is to find a solution S^* such that the *expected value* of $PM(S^*)$ is maximum.

Furthermore, two additional secondary objectives are considered as well: minimizing the expected total delay, and minimizing the expected number of delayed passengers. With regard to the on-time objective, the importance of the total delay magnitude (and not only the number of delays) is clear since large delays would inevitably translate into a stream of reactionary delays for all downstream flights. This cascading effect, which is often referred to as the snowball effect, has a devastating impact on airline schedule adherence. On the other hand, the importance of the third objective stems from the fact that the number of delayed passengers directly impinges on the airline's reputation and on the delay costs, which include the cost of accommodating delayed passengers and rerouting passengers that miss their connections.

Hence, the RWARR is modeled as a stochastic multi-objective combinatorial optimization problem.

Remark 1. The model that is addressed in this paper differs in many aspects from the robust weekly aircraft maintenance routing problem that is investigated in Liang et al. (2015). Firstly, these authors did not consider the retiming option for flight departure times, but focused on the aircraft routing problem only. Therefore, the problem they addressed does not include the complicating synchronization constraints, and thereby can be decomposed into independent problems for the different fleet types. Also, they considered the objective of minimizing the sum of the expected reactionary delays (and not the on-time performance). Since the time complexity for computing the expected reactionary delays for an aircraft route is an exponential function of the number of flights that are served on this route, the authors assumed that all aircraft should remain idle overnight for a few hours. This assumption drastically restricts the number of flights that are included in an LOF to 5 or 6. In our case, we consider a general flight schedule including a mix of both long- and short-haul flights. Therefore, some aircraft routes may include up to 30 flights, or even more, with no idle overnight time (which makes it intractable to accurately assess the associated expected reactionary delays). Finally, the authors proposed a formulation having exponentially many variables, which was solved using a sophisticated two-stage column generation algorithm. In contrast, we model the robust weekly aircraft maintenance routing problem using a compact formulation (described in Section 4.1.) which can be directly solved using a commercial MIP solver.

3.2. Overview of the solution strategy

Prior to describing the proposed solution strategy, it is worth providing some insights that may shed light on the problem difficulty. Toward this end, and for the sake of simplicity, we tentatively assume that the routing decisions have been already made and that the problem requires setting the final departure times with the objective of maximizing the on-time performance. To avoid making the analysis unnecessarily complicated, we shall impose two simplifying assumptions. First, we shall relax the synchronization constraint that requires that all flights within the same cluster should be assigned the same departure time. Second, we shall analyze a single aircraft route. Furthermore, we assume that the analytical forms of the distribution functions of the primary delays are given.

Data notation

$\sigma = (\sigma(1), \ldots, \sigma(nf))$ circular sequence of nf flights that are consecutively flown by an aircraft. In what follows we set $\sigma(0) \equiv \sigma(nf)$ and define,
$e_{\sigma(j)}/l_{\sigma(j)}$ earliest/latest departure time of flight $\sigma(j),\ j = 1, \ldots, nf$;
$\varphi_{\sigma(j)}$ sum of the duration of flight $\sigma(j)$ and the aircraft turn time, $j = 1, \ldots, nf$;
$p_{\sigma(j)}$ probability of having a nonzero primary delay for flight $\sigma(j),\ j = 1, \ldots, nf$;
$d_{\sigma(j)}$ random variable that represents the nonzero primary delay of flight $\sigma(j)$ $(d_{\sigma(j)} > 0),\ j = 1, \ldots, nf$;
$g_{\sigma(j)}(.)$ probability density function of $d_{\sigma(j)},\ j = 1, \ldots, nf$.

For the sake of tractability, we make the following assumptions.

Assumption 1. A right-shift (or push-back) recovery strategy is used.

Assumption 2. The primary delays of the sequence of flights that are served by an aircraft are independent of each other.

The first assumption requires that a flight is delayed until the aircraft becomes available. The second assumption requires taking no account of primary delay causes (e.g. inclement weather or aircraft failure) that may simultaneously impact consecutive flights (see Yan and Kung, 2016).

Additional notation

$\bar{t}_{\sigma(j)}$ planned departure time of flight $\sigma(j), j = 1, \ldots, nf$;
$\Delta_{\sigma(j)} = \bar{t}_{\sigma(j)} - (\bar{t}_{\sigma(j-1)} + \varphi_{\sigma(j-1)})$ planned slack of flight $\sigma(j),\ j = 1, \ldots, nf$; i.e., difference between the departure time of $\sigma(j)$ and the arrival time of flight $\sigma(j-1),\ j = 1, \ldots, nf$;
$\pi_{\sigma(j)}$ probability that flight $\sigma(j)$ will experience a nonzero delay, $j = 1, \ldots, nf$;
$\delta_{\sigma(j)}$ random variable that represents the total nonzero delay of flight $\sigma(j)$ $(\delta_{\sigma(j)} > 0),\ j = 1, \ldots, nf$; (this delay is the sum of the primary delay of $\sigma(j)$ and the reactionary delay),
$f_{\sigma(j)}(.)$ Probability density function of the total nonzero delay of flight $\sigma(j),\ j = 1, \ldots, nf$;
$F_{\sigma(j)}(.)$ Cumulative distribution function of the total nonzero delay of flight $\sigma(j),\ j = 1, \ldots, nf$;
$t_{\sigma(j)}$ realized departure time of flight $\sigma(j)$ (that is, $t_{\sigma(j)} = \bar{t}_{\sigma(j)} + \delta_{\sigma(j)}$, if $\sigma(j)$ is delayed, and $t_{\sigma(j)} = \bar{t}_{\sigma(j)}$ otherwise), $j = 1, \ldots, nf$.

Claim 1. *The probability of having a nonzero delay for flight* $\sigma(j)$ *is given by*

$$\pi_{\sigma(j)} = 1 - (1 - p_{\sigma(j)})((1 - \pi_{\sigma(j-1)}) + \pi_{\sigma(j-1)} F_{\sigma(j-1)}(\Delta_{\sigma(j)})), \tag{1}$$

for all $j = 1, \ldots, nf$.

Proof. Flight $\sigma(j)$ is not delayed if and only if both the following events occur:

- There is no primary delay for flight $\sigma(j)$. The probability of this event is $1 - p_{\sigma(j)}$.
- There is no delay for flight $\sigma(j-1)$ (the probability of this event is $1 - \pi_{\sigma(j-1)}$), or there is a delay for flight $\sigma(j-1)$ and this delay is shorter than the planned slack $\Delta_{\sigma(j)}$ (the probability of this event is $\pi_{\sigma(j-1)}F_{\sigma(j-1)}(\Delta_{\sigma(j)})$).

Hence, we have

$$1 - \pi_{\sigma(j)} = (1 - p_{\sigma(j)})((1 - \pi_{\sigma(j-1)}) + \pi_{\sigma(j-1)}F_{\sigma(j-1)}(\Delta_{\sigma(j)})). \quad \square$$

Claim 2. *The probability density function of the total delay of flight $\sigma(j)$ is given by*

$$f_{\sigma(j)}(t) = \alpha_{\sigma(j)} \int_{\Delta_{\sigma(j)}}^{t+\Delta_{\sigma(j)}} g_{\sigma(j)}(t - y + \Delta_{\sigma(j)}) f_{\sigma(j-1)}(y)dy + \beta_{\sigma(j)} f_{\sigma(j-1)}(t + \Delta_{\sigma(j)}) + \gamma_{\sigma(j)} g_{\sigma(j)}(t)dy,$$

where

$$\begin{aligned} \alpha_{\sigma(j)} &= p_{\sigma(j)}\pi_{\sigma(j-1)}(1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})) \\ \beta_{\sigma(j)} &= \pi_{\sigma(j-1)}(1 - p_{\sigma(j)})(1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})) \qquad \text{for all } j = 1, \ldots, nf. \\ \gamma_{\sigma(j)} &= p_{\sigma(j)}((1 - \pi_{\sigma(j-1)}) + \pi_{\sigma(j-1)}F_{\sigma(j-1)}(\Delta_{\sigma(j)})) \end{aligned}$$

Proof. If flight $\sigma(j)$ is delayed then we have $t_{\sigma(j)} = \bar{t}_{\sigma(j)} + \delta_{\sigma(j)}$. The delay $\delta_{\sigma(j)}$ can result from: (i) the sum of a primary delay of flight $\sigma(j)$ and a reactionary delay from flight $\sigma(j-1)$, (ii) the reactionary delay from flight $\sigma(j-1)$ only, or (iii) the primary delay of flight $\sigma(j)$ only. $\square$

Case (i): This case occurs when: (a) there is a primary delay for flight $\sigma(j)$ (probability = $p_{\sigma(j)}$), and (b) there is a delay for flight $\sigma(j-1)$ (probability = $\pi_{\sigma(j-1)}$), and (c) the delay of flight $\sigma(j-1)$ is larger than the slack $\Delta_{\sigma(j)}$ (probability = $1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})$). Hence, we have

$$\delta_{\sigma(j)} = d_{\sigma(j)} + \delta_{\sigma(j-1)} - \Delta_{\sigma(j)}.$$

Therefore, Case (i) will occur with a probability

$$\alpha_{\sigma(j)} = p_{\sigma(j)}\pi_{\sigma(j-1)}(1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})),$$

and has a corresponding probability density function (PDF)

$$f_{\sigma(j)}(t) = \int_{\Delta_{\sigma(j)}}^{t+\Delta_{\sigma(j)}} g_{\sigma(j)}(t - y + \Delta_{\sigma(j)}) f_{\sigma(j-1)}(y)dy.$$

Case (ii): This case occurs when: (a) there is no primary delay for flight $\sigma(j)$ (probability = $1 - p_{\sigma(j)}$), (b) there is a delay for flight $\sigma(j-1)$ (probability = $\pi_{\sigma(j-1)}$), and (c) the delay of flight $\sigma(j-1)$ is larger than the slack $\Delta_{\sigma(j)}$ (probability = $1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})$). Hence, we have

$$\delta_{\sigma(j)} = \delta_{\sigma(j-1)} - \Delta_{\sigma(j)}.$$

Therefore, Case (ii) will occur with a probability

$$\beta_{\sigma(j)} = (1 - p_{\sigma(j)})\pi_{\sigma(j-1)}(1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})),$$

and has an associated PDF

$$f_{\sigma(j)}(t) = f_{\sigma(j-1)}(t + \Delta_{\sigma(j)}).$$

Case (iii): This case occurs when: (a) there is a primary delay for flight $\sigma(j)$ (probability = $p_{\sigma(j)}$), and (b) there is no delay for flight $\sigma(j-1)$ (probability = $1 - \pi_{\sigma(j-1)}$), or (c) there is a delay of flight $\sigma(j-1)$ and the delay is smaller than the slack $\Delta_{\sigma(j)}$ (probability = $\pi_{\sigma(j-1)}F_{\sigma(j-1)}(\Delta_{\sigma(j)})$). Hence, we have

$$\delta_{\sigma(j)} = d_{\sigma(j)}.$$

Case (iii) will occur with a probability

$$\gamma_{\sigma(j)} = p_{\sigma(j)}((1 - \pi_{\sigma(j-1)}) + \pi_{\sigma(j-1)}F_{\sigma(j-1)}(\Delta_{\sigma(j)})),$$

and has an associated PDF

$$f_{\sigma(j)}(t) = g_{\sigma(j)}(t).$$

Therefore, we derive the PDF of the delay of flight $\sigma(j)$

$$f_{\sigma(j)}(t) = \alpha_{\sigma(j)} \int_{\Delta_{\sigma(j)}}^{t+\Delta_{\sigma(j)}} g_{\sigma(j)}(t - y + \Delta_{\sigma(j)}) f_{\sigma(j-1)}(y)dy + \beta_{\sigma(j)} f_{\sigma(j-1)}(t + \Delta_{\sigma(j)}) + \gamma_{\sigma(j)} g_{\sigma(j)}(t)dy,$$

Remark 2. Given $\Delta_{\sigma(j)}$ $(j = 1, \ldots, nf)$, the computation of $\pi_{\sigma(j)}$ is cumbersome because of the recursive definition of these variables. Indeed, the value of $\pi_{\sigma(j)}$ depends on $\pi_{\sigma(j-1)}$ and (by recursion) the value of this latter entity depends on $\pi_{\sigma(j)}$. This could be illustrated by the following special case. Assume that $\sigma(n)$ exhibits a large buffer time that satisfies $F_{\sigma(n-1)}(\Delta_{\sigma(n)}) = 1$, and that buffer times of all other flights are equal to zero. That is, we have $F_{\sigma(j-1)}(\Delta_{\sigma(j)}) = 0$, for all $j = 1, \ldots n-1$. In this case, (1) yields

$$\pi_{\sigma(n)} = p_{\sigma(n)}$$

and

$$\pi_{\sigma(j)} = 1 - (1 - p_{\sigma(j)})(1 - \pi_{\sigma(j-1)}) \quad \text{for } j = 1, \ldots, n-1.$$

Hence, we see that in this case, we have

$$\pi_{\sigma(n-1)} = 1 - (1 - \pi_{\sigma(n)}) \prod_{l=1}^{n-1} (1 - p_{\sigma(l)}).$$

The on-time performance of flight $\sigma(j)$ is equal to the non-delay probability $1 - \pi_{\sigma(j)}$. Hence, it follows that an estimate of the on-time performance of the aircraft route is $1 - \sum_{j=1}^{nf} \pi_{\sigma(j)}/nf$.

Therefore, maximizing the on-time performance of the aircraft route requires solving the following retiming problem:

$$(OTP): \text{ Minimize } \sum_{j=1}^{nf} \pi_{\sigma(j)} \tag{2}$$

subject to:

$$\pi_{\sigma(j)} + (1 - p_{\sigma(j)})((1 - \pi_{\sigma(j-1)}) + \pi_{\sigma(j-1)} F_{\sigma(j-1)}(\Delta_{\sigma(j)})) = 1, \quad j = 1, \ldots, nf, \tag{3}$$

$$p_{\sigma(j)} \pi_{\sigma(j-1)} (1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})) - \alpha_{\sigma(j)} = 0, \quad j = 1, \ldots, nf, \tag{4}$$

$$\pi_{\sigma(j-1)} (1 - p_{\sigma(j)})(1 - F_{\sigma(j-1)}(\Delta_{\sigma(j)})) - \beta_{\sigma(j)} = 0, \quad j = 1, \ldots, nf, \tag{5}$$

$$p_{\sigma(j)}((1 - \pi_{\sigma(j-1)}) + \pi_{\sigma(j-1)} F_{\sigma(j-1)}(\Delta_{\sigma(j)})) - \gamma_{\sigma(j)} = 0, \quad j = 1, \ldots, nf, \tag{6}$$

$$\begin{aligned} &\alpha_{\sigma(j)} \int_{\Delta_{\sigma(j)}}^{t+\Delta_{\sigma(j)}} g_{\sigma(j)}(t - y + \Delta_{\sigma(j)}) f_{\sigma(j-1)}(y)dy + \beta_{\sigma(j)} f_{\sigma(j-1)}(t + \Delta_{\sigma(j)}) \\ &\quad + \gamma_{\sigma(j)} g_{\sigma(j)}(t) = f_{\sigma(j)}(t), \quad j = 1, \ldots, nf, \end{aligned} \tag{7}$$

$$F_{\sigma(j-1)}(\Delta_{\sigma(j)}) - \int_0^{\Delta_{\sigma(j)}} f_{\sigma(j-1)}(t)dt = 0, \quad j = 1, \ldots, nf, \tag{8}$$

$$t_{\sigma(j)} - t_{\sigma(j-1)} - \Delta_{\sigma(j)} = \varphi_{\sigma(j-1)}, \quad j = 1, \ldots, nf, \tag{9}$$

$$\Delta_{\sigma(j)} \geqslant 0, \quad j = 1, \ldots, nf, \tag{10}$$

$$e_{\sigma(j)} \leqslant t_{\sigma(j)} \leqslant l_{\sigma(j)}, \quad j = 1, \ldots, nf. \tag{11}$$

The objective function (2) minimizes the sum of nonzero delay probabilities (which is equivalent to maximizing the on-time performance). Constraint (3) relates the probabilities of having a nonzero delay for pairs of consecutive flights. Equalities Eqs. (4)–(6) express the occurrence probabilities of Cases (i), (ii), and (iii), respectively. Equalities (7) and (8) compute the PDF and CDF of the delay of each flight, respectively. Constraint (9) together with (10) enforce the relationship between the time slacks and the departure times. Finally, (11) are the time-window constraints.

Remark 3. Define $r_{\sigma(j)}$ as the probability of having a nonzero reactionary delay for flight $\sigma(j)$ $(j = 1, \ldots, nf)$. The probability that flight $\sigma(j)$ will not experience any delay is given by

$$1 - \pi_{\sigma(j)} = (1 - p_{\sigma(j)})(1 - r_{\sigma(j)}).$$

Thus, we have

$$\pi_{\sigma(j)} = p_{\sigma(j)} + (1 - p_{\sigma(j)}) r_{\sigma(j)}, \quad \text{for } j = 1, \ldots, nf.$$

Hence, we see that the objective function (2) amounts to minimizing a weighted sum of the reactionary delays (that is, we seek to *Minimize* $\sum_{j=1}^{nf} (1 - p_{\sigma(j)}) r_{\sigma(j)}$).

Clearly, the exact solution of the retiming problem (OTP) is (generally) a daunting task. Furthermore, recall that the problem also requires synchronizing the departure times within each cluster, and, even more difficult, solving an embedded maintenance aircraft routing problem, which is an $\mathcal{NP}$-hard problem (Talluri, 1998 b).

In Sections 4 and 5, we shall propose an approximate solution strategy that requires decomposing the problem into two subproblems that are sequentially solved using the following two-stage optimization-simulation approach:

- **Stage 1: Construction of maintenance feasible aircraft routes**. We fix the flight departure times at their earliest feasible departure times, and we formulate the weekly aircraft routing problem as a compact nonlinear mixed-integer program with a tailored surrogate objective function. An equivalent linearized model is then derived using the Reformulation Linearization Technique of Sherali and Adams (1990, 1994).
- **Stage 2: Computation of the departure times**. Given the aircraft routes that are constructed in Stage 1, we heuristically solve a retiming problem in the spirit of Model *OTP*, while additionally accommodating the synchronization constraint. Toward this end, we implement an adjustment heuristic that requires to iteratively invoke a Monte-Carlo simulation procedure for estimating the expected delays and adjusting the slacks to improve the on-time performance.

4. An exact model for the robust aircraft maintenance routing problem with fixed departure times

In Stage 1, we first assume that for each flight cluster the departure times are set at their respective earliest departure times. Note that the first-stage problem is decomposable into K independent problems (one for each aircraft type). In this section, we shall describe the model in two steps. First, we shall investigate the problem of solving the feasibility variant of the weekly aircraft maintenance routing problem. Next, we shall describe the surrogate robustness objective function.

4.1. A compact formulation for the weekly aircraft maintenance routing problem

To model the weekly aircraft maintenance routing problem that is associated with aircraft type k, we define an associated digraph $G_k = (V_k, A_k)$ in which each node $j \in V_k$ represents a flight leg that should be flown by an aircraft of type k, and where an arc $(i,j) \in A_k$ if and only if an aircraft of type k can be successively assigned to legs i and j. Following Haouari et al. (2013), who addressed the *daily* aircraft maintenance routing problem, we define the arcset A_k as the union of four arc subsets A_1^k, A_2^k, A_3^k, and A_4^k that are defined as follows:

- An arc $(i,j) \in A_1^k$ if and only if a maintenance check could be planned between the arrival of flight i and the departure of flight j, and both flights are consecutively flown on the same day. Hence, $(i,j) \in A_1^k \iff (1) AS_i \equiv DS_j$, $(2)\ AS_i \in S$; $(3)\ DO_i = DO_j$, and $(4)\ AT_i + M_k \leqslant DT_j$.
- An arc $(i,j) \in A_2^k$ if and only if a maintenance check could be planned between the arrival of flight i and the departure of flight j, and both flights are consecutively flown on two consecutive days. Hence, $(i,j) \in A_2^k \iff (1) AS_i \equiv DS_j$, $(2) AS_i \in S$; (3) $DO_j = DO_i + 1 \pmod 7$; and $(4)\ AT_i + M_k \leqslant DT_j$ if $DO_i \leqslant 6$, and $AT_i + M_k \leqslant DT_j + 10{,}080$ if $DO_i = 7$.
- An arc $(i,j) \in A_3^k$ if and only if a maintenance check could not be planned between the arrival of flight i and the departure of flight j, and both flights are flown consecutively on the same day. Hence, $(i,j) \in A_3^k \iff (1) AS_i \equiv DS_j$, $(2)\ AS_i \notin S$ or $DT_j < AT_i + M_k$; $(3)\ DO_i = DO_j$, and $(4)\ AT_i + \theta_k \leqslant DT_j$.
- An arc $(i,j) \in A_4^k$ if and only if a maintenance check could not be planned between the arrival of flight i and the departure of flight j, and both flights are consecutively flown on two consecutive days. Hence, $(i,j) \in A_4^k \iff (1) AS_i \equiv DS_j$; $(2) AS_i \notin S$, or $(DT_j < AT_i + M_k$ and $DO_i \leqslant 6)$, or $(DT_j + 10{,}080 < AT_i + M_k$ and $DO_i = 7)$, (3) $DO_j = DO_i + 1 \pmod 7$; and (4) $(AT_i + \theta_k \leqslant DT_j$ and $DO_i \leqslant 6)$, or $(AT_i + \theta_k \leqslant DT_j + 10{,}080$ and $DO_i = 7)$.

Remark 4. For the sake of computational convenience, and in alignment with common airlines practices, we drop from A_k all connections whose corresponding aircraft idle time is larger than 24 h.

The arc subset $A_M^k \equiv A_1^k \cup A_2^k$ shall be referred to as the maintenance arcset. It includes all *maintenance connections* during which a maintenance check could be scheduled. Similarly, $A_{\overline{M}}^k = A_3^k \cup A_4^k$ corresponds to the set of non-maintenance connections.

Furthermore, an arc $(i,j) \in A$ is referred to as a *wraparound arc* (or *connection*) if flight i departs on a given day of the week and j departs on a day of the next week. That is, (i,j) is a wraparound connection if either flight i departs on a given day, crosses the end of the planning horizon, and arrives on the next week, or an aircraft remains grounded, after serving i and before serving j, at midnight on the last day of the week. We denote by A_w^k the set of wraparound connections.

Now, consider a cycle $R = (j_1, j_2, \ldots, j_p, j_1)$ in G_k, where Rcorresponds to an aircraft rotation that consecutively serves flights legs $j_1, j_2, \ldots, j_p$ and returns to j_1 in a cyclic fashion. We make the following observation.

Proposition 1. *If R includes w wraparound connections then it spans w weeks. Therefore, R should be covered by w identical aircraft.*

Proof. This is an immediate consequence of the definitions of wraparound connections, and of the weekly periodicity of the flight schedule. □

Using these definitions, we can formulate the problem of finding a feasible solution for the weekly aircraft maintenance routing problem as follows.

Decision variables

- x_{ij}: binary variable that takes 1 if connection $(i,j) \in A_k$ is served by an aircraft, and 0 otherwise.
- m_j: number of elapsed days for an aircraft since the last maintenance check after serving leg $j \in V_k$.

The feasibility variant of weekly aircraft maintenance routing problem reads as follows:

$$\text{Find } x, m \tag{12}$$

subject to

$$\sum_{i:(i,j)\in A_k} x_{ij} = 1, \quad \forall j \in V_k, \tag{13}$$

$$\sum_{i:(j,i)\in A_k} x_{ji} = 1, \forall j \in V_k, \tag{14}$$

$$m_j x_{ij} = x_{ij}, \quad \forall j \in V_k,\ (i,j) \in A_M^k, \tag{15}$$

$$m_j x_{ij} = m_i x_{ij}, \quad \forall j \in V_k,\ (i,j) \in A_3^k, \tag{16}$$

$$m_j x_{ij} = (m_i + 1) x_{ij}, \quad \forall j \in V_k, (i,j) \in A_4^k \tag{17}$$

$$\sum_{(i,j)\in A_w^k} x_{ij} \leqslant NA_k, \tag{18}$$

$$1 \leqslant m_j \leqslant \Omega_k, \quad \forall j \in V_k, \tag{19}$$

$$x_{ij} \in \{0,1\}, \quad \forall (i,j) \in A_k. \tag{20}$$

Constraints (13) and (14) ensure that each flight has exactly one predecessor and one successor, respectively. The *nonlinear* Constraints Eqs. (15)–(17) and (19) enforce the restriction on the maximum numbers of days between consecutive maintenance checks. Note that, given binary values of x feasible to Constraints (15)–(17), Constraints (15)–(17) necessarily induce the m-variables to be integer-valued. Constraints (18) requires that the total number of aircraft in service should not exceed the available fleet size, and Constraints (19) and (20) represent logical restrictions.

4.2. Model linearization

To enhance the model solvability, we derive an equivalent mixed-integer linear programming model by applying the Reformulation Linearization Technique of Sherali and Adams (1990, 1994).

To begin with, consider Constraints (15)–(17). We can linearize these constraints using the following substitutions:

$$\lambda_{jl} = m_j x_{jl}, \quad \forall (j,l) \in A_k, \tag{21}$$

$$\mu_{jl} = m_l x_{jl}, \quad \forall (j,l) \in A_k. \tag{22}$$

Thus, (15)–(17) can be transformed to the following:

$$\mu_{jl} = x_{jl}, \quad \forall (j,l) \in A_M^k, \tag{23}$$

$$\mu_{jl} = \lambda_{jl}, \quad \forall (j,l) \in A_3^k, \tag{24}$$

$$\mu_{jl} = \lambda_{jl} + x_{jl}, \quad \forall (j,l) \in A_4^k. \tag{25}$$

Also, multiplying (19) by each x_{lj} and x_{jl}, respectively, and linearizing using (21) and (22), we obtain (after rearranging indices):

$$x_{jl} \leqslant \lambda_{jl} \leqslant \Omega_k x_{jl}, \quad \forall (j,l) \in A_k \tag{26}$$

and

$$x_{jl} \leqslant \mu_{lj} \leqslant \Omega_k x_{jl}, \quad \forall\ (j,l) \in A_k \tag{27}$$

Next, multiplying (13) and (14) by its corresponding m_j and linearizing, we obtain:

$$\sum_{l:(l,j)\in A} \mu_{lj} = m_j, \quad \forall j \in V_k, \tag{28}$$

$$\sum_{l:(l,j)\in A} \lambda_{jl} = m_j, \quad \forall j \in V_k, \tag{29}$$

Proposition 2. *Constraints (23)–(29) can be replaced by the following restrictions (30)–(32), which, together with (14), are equivalent to these constraints even in the continuous (LP relaxation) sense:*

$$\sum_{l:(j,l)\in A} \lambda_{jl} = \sum_{l:(l,j)\in A_M \cup A_4} x_{lj} + \sum_{l:(l,j)\in A_{\overline{M}}} \lambda_{lj}, \quad \forall j \in V_k, \tag{30}$$

$$x_{jl} \leqslant \lambda_{jl} \leqslant (\Omega_k - 1)x_{jl}, \quad \forall (j,l) \in A_4, \tag{31}$$

$$x_{jl} \leqslant \lambda_{jl} \leqslant \Omega_k x_{jl}, \quad \forall (j,l) \in A_M \cup A_3, \tag{32}$$

Proof. Using the identities (28) and (29), we get

$$\sum_{l:(l,j)\in A} \mu_{jl} = \sum_{l:(l,j)\in A} \lambda_{jl}, \quad \forall j \in V_k. \quad \square$$

Next, we eliminate the μ-variables from these restrictions to derive the required result.

4.3. Surrogate robustness measure

In what follows, we define the slack s_{ij} of $(i,j) \in A$ as the difference between the departure time of flight j and the aircraft ready time. That is, we have

$$s_{ij} \equiv DT_j - AT_i - \theta_k, \quad \text{if } DT_j \geqslant AT_i + \theta_k, \quad \text{and} \quad DT_j + 10,080 - AT_i - \theta_k, \quad \text{otherwise.}$$

Inserting a slack between two consecutive flights often enhances the connection's local robustness since this slack usefully acts as a cushion against unplanned delays. However, imposing large slacks (for example, when the slack is much larger than the 95th percentile delay) may prove inefficient and result in low aircraft utilization. On the other hand, setting too short slacks may seriously harm the schedule's robustness. Thus, to promote the generation of robust aircraft routes, we define a surrogate objective function, in the same vein as in Aloulou et al. (2013), which aims at judiciously inserting buffer times (i.e., slacks) between connecting flights. Toward this end, we define for each connection (i,j) a continuous utility function $u(.)$ that maps the slack s into a real number in $[0,1]$ that expresses the slack's ability to absorb reactionary delays. We postulate that the slack utility function should exhibit the following properties:

1. Nondecreasing for $s \geqslant 0$,
2. Diminishing marginal returns for $s \geqslant t$.

Property 1 states that increasing the slack cannot deteriorate its ability to absorb reactionary delays. Property 2 states that if the slack exceeds some threshold value t, then the marginal utility should be decreasing (or equivalently, the rate at which the utility increases begins to slow). This property is motivated by the following saturation effect: elongating large buffers has a negligible impact on the connection's robustness.

Many utility functions that satisfy Properties 1 and 2 could be implemented. In the present research, we illustrate our hybrid simulation–optimization approach by considering the following function:

$$u(s) = \frac{s}{s+\beta} \quad \text{for } s > 0,$$

where β is a strictly positive parameter that is set empirically (see Section 6.3).

It is easy to check that $u(.)$ satisfies both Properties 1 and 2. Indeed, we have.

- $\frac{\partial u}{\partial s}(s) = \frac{\beta}{(s+\beta)^2} > 0$ **for** $s \geqslant 0$,
- $\frac{\partial^2 u}{\partial s^2}(s) = -2\frac{\beta}{(s+\beta)^3} < 0$ **for** $s \geqslant 0$.

Hence, a surrogate measure of the local robustness of connection $(i,j) \in A$ is $\gamma_{ij} \equiv u(s_{ij})$.

Accordingly, the aircraft maintenance routing problem can be stated as a mixed-integer linear program as follows:

$$(MILP): \ \text{Maximize} \sum_{(i,j)\in A_k} \gamma_{ij} x_{ij} \tag{33}$$

subject to

$$\sum_{i\in\delta_j^-} x_{ij} = 1, \quad \forall j \in V_k, \tag{34}$$

$$\sum_{i\in\delta_j^+} x_{ji} = 1, \quad \forall j \in V_k, \tag{35}$$

$$\sum_{i:(j,i)\in A} \lambda_{ji} = 1 + \sum_{i:(j,i)\in A_{\overline{M}}} \lambda_{ji} - \sum_{i:(j,i)\in A_3} x_{ji}, \quad \forall j \in V_k, \tag{36}$$

$$x_{ij} \leqslant \lambda_{ij} \leqslant b_{ij}^k x_{ij}, \quad \forall (i,j) \in A_k, \tag{37}$$

$$\sum_{(i,j)\in A_w^k} x_{ij} \leqslant NA_k, \tag{38}$$

$$x_{ij} \in \{0,1\}, \quad \forall (i,j) \in A_k. \tag{39}$$

where for the sake of convenience, b_{ij}^k is defined as follows:

$$b_{ij}^k = \begin{cases} \Omega_k - 1, & \forall (i,j) \in A_4^k, \\ \Omega_k, & \forall (i,j) \in A_k \setminus A_4^k. \end{cases}$$

Note that here again only x-variables are explicitly restricted to be integer-valued and that the λ-variables are declared continuous, since they will automatically take on integer values for any feasible solution upon using (29).

5. A simulation-based approach for flight retiming

In Stage 1 discussed above, we assumed that all the flight cluster departure times are fixed at their earliest departure times, and we derived maintenance feasible aircraft routes by solving Model *MILP*. In Stage 2, we now address the issue of revising the flight cluster departure times with the objective of enhancing the robustness of the overall solution. That is, we compute the final flight departure times while accommodating the following restrictions:

1. Synchronization constraints: the flights that belong to the same cluster should depart at the same time of the day.
2. Time windows constraints: each flight departure time should lie within the specified time window.
3. Routing constraints: the aircraft routes that were obtained upon Stage 1 should not be altered. Furthermore, maintenance feasibility should be preserved.

For this purpose, we implemented an iterative heuristic approach that alternates between a Monte-Carlo simulation procedure (for estimating the delays), and a retiming algorithm (for increasing the buffer times of the most vulnerable connections that exhibit large reactionary delays). Using Remark 3, we observe that an effective approach for flight retiming should focus on those critical connections that exhibit large reactionary delays. Indeed, the insertion of buffer times may prove useful for mitigating reactionary delays only since primary delays are not dependent on buffer times.

The underlying idea of the heuristic is to adopt Monte-Carlo simulation for estimating the reactionary delays of all flights and then use this information for enhancing the solution robustness. More precisely, assume that a cluster c, which is not departing at its latest feasible time, exhibits a large weighted reactionary delay (as ascribed by Remark 3), while its immediate downstream flights exhibit a much smaller weighted reactionary delays. Then the departure times of the flights that belong to $c$can be right-shifted (i.e., increased) by a small time-step ts. In so doing, we increase the size of each buffer time that is planned for a flight in cluster c by ts units and thereby enhance the on-time performance of these flights. Furthermore, the buffer times that are planned for flights following those in cluster c are shortened by ts units of time. However, if these succeeding flights exhibit negligible reactionary delays, then the overall outcome of this time adjustment will prove positive. Similarly, we may use the information on the realized reactionary delays for promoting the solution's on-time performance through left-shifting, i.e., when feasible, decreasing the departure times of some clusters. Indeed, if cluster c that is not departing at its earliest feasible time exhibits a much smaller cumulative reactionary delay than its immediate downstream flights, then the departure times of the flights that belong to c can be usefully left-shifted by a time-step ts.

Prior to providing a detailed description of the heuristic, we provide some additional notation.

- τ_c: departure time of flight cluster $c, c = 1, \ldots, C$.
- $\tau_c^{\min}/\tau_c^{\max}$: earliest/latest feasible departure time of flight cluster $c, c = 1, \ldots, C$. These bounds are derived from the time windows of the flights that are included in c, as well as the planned maintenance periods that are ascribed by the aircraft routes that are built in Stage 1.
- $\tilde{r}_j$: estimate of the probability of nonzero reactionary delay for flight $j \in L$ (see Remark 3).
- $s_c \equiv \sum_{j\in c}(1-p_j)\tilde{r}_j$: weighted sum of the estimates of the probabilities of reactionary delays of all flights that belong to cluster c, $c = 1, \ldots, C$.

- $s_c^+ \equiv \sum_{j \text{ successor of } c}(1-p_j)\tilde{r}_j$: weighted sum of the estimates of the probabilities of reactionary delays of all flights that are immediate successors of flights that belong to cluster c, $c = 1,\ldots,C$.
- PM: percentage of flights that were not delayed by more than 15 min.

The proposed simulation procedure used for assessing the solution performance is described below.

Monte-Carlo simulation procedure
Input: Set of aircraft routes and corresponding departure times.
For each aircraft route in the solution, Repeat N cycles of Steps (a) and (b):

Step (a) - Randomly generate primary delays.

Step (b) - Propagate the delays through the aircraft rotation, and compute the realized reactionary delays.

Step (c) - Compute, for each flight j, an estimate of the probability $\tilde{r}_j$ of occurrence of a reactionary delay.

Step (d) - Compute s_c and s_c^+ for $c = 1,\ldots,C$.

Step (e) - Compute the on-time performance PM.

The main steps of the retiming heuristic are described next:

Retiming procedure
Phase I: **Right-shift of the departure times**
Step 1 - **Initialization**. Set $\tau_c \leftarrow \tau_c^{\min}$ for $c = 1,\ldots,C, PM_{best} = 0$.
Step 2 - Invoke the Monte-Carlo simulation procedure.
Step 3 - Retiming of selected clusters:
(3.1) - If $PM > PM_{best}$ then set $PM_{best} \leftarrow PM$. Otherwise, if PM has not improved during $iter_{\max}$ consecutive iterations then go to Step 4.
(3.2) - Set $\Omega^+ = \{c = 1,\ldots,C : s_c \geqslant \epsilon_1 s_c^+ \text{ and } \tau_c \leqslant \tau_c^{\max} - ts\}$.
(3.3) - If $\Omega^+ = \emptyset$then go to Step 4.
(3.4) - Set $\tau_c \leftarrow \tau_c + ts$, for all $c \in \Omega^+$, and go to Step 2.
Phase II: **Left-shift of the departure times**
Step 4 - Retiming of selected clusters:
(4.1) - Set $\Omega^- = \{c = 1,\ldots,C : s_c \leqslant \epsilon_2 s_c^+ \text{ and } \tau_c \geqslant \tau_c^{\min} + ts\}$.
(4.2) - If $\Omega^- = \emptyset$ then **Stop**.
(4.3) - Set $\tau_c \leftarrow \tau_c - ts$, for all $c \in \Omega^-$.
Step 5 - Invoke the Monte-Carlo simulation procedure.
Step 6 - If $PM > PM_{best}$ then set $PM_{best} \leftarrow PM$ and go to Step 4.
Step 7 - If PM has not improved during $iter_{\max}$ consecutive iterations then **Stop**; else go to Step 4.1.

The final solution corresponds to the set of departure times that yield the highest on-time performance PM_{best}.

In our implementation, we empirically set the parameters as follows: time-step $ts = 5$, number of simulation runs $N = 1000$, maximum number of non-improving iterations $iter_{\max} = 5$ consecutive iterations, $\epsilon_1 = 2$, and $\epsilon_2 = 0.5$.

6. Computational experiments

To assess the performance of the proposed two-stage approach, we conducted a computational study using *SimAir*, which is a simulation tool for assessing airline operations with disruptions. In this section, we provide an overview of *SimAir*. Next, we describe our set of instances. Finally, we will present comparative results obtained from *SimAir*, which provide evidence of the robustness of solutions produced by our approach.

6.1. SimAir: a simulation of airline operations

SimAir is a modular simulation tool for evaluating plans and recovery procedures. It is currently being used by major airline companies and academic researchers. A detailed description of this tool and its structure is given in Rosenberger et al. (2002). We used *SimAir* to study aircraft plans and passenger itineraries under a specific recovery method and delay distribution. *SimAir* provides preset empirical distributions to describe a series of scenarios of disruptions such as gate delay probabilities, aircraft take-off and landing durations, taxi-out and taxi-in duration, and probability and duration of unscheduled maintenance. By manipulating this set of stochastic events, disruption scenarios are simulated into the flight schedule over multiple iterations. In the aftermath of each disruption, a recovery technique is invoked with actions to bring the schedule back to feasibility. Though *SimAir* provides a pool of recovery techniques, we focus throughout this work on the *push-back* and *flight cancellation* techniques since they are often used in practice. Alternative techniques may include diverting aircraft in the air, putting legs on hold, ferrying of aircraft to stations with maintenance capability to undergo maintenance tasks, etc. Finally, we extracted the following measures of performance to evaluate the robustness of our solutions:

- PM_1: The schedule's overall on-time performance (delays less than 15 min).
- PM_2: The total amount of delays that exceed 15 min.
- PM_3: The total number of delayed passengers.

6.2. Description of problem instances

We considered three data instances, hereafter referred to by QA_1, QA_2 and QA_3, with the number of flight legs ranging between 2906 and 3387, and the number of aircraft ranging between 141 and 164. For all instances, the number of aircraft types is $K = 10$ and include a mix of wide- and narrow-body aircraft. These instances correspond to different one-week schedules that were operated by Qatar Airways at different periods of the year. The main characteristics of the three instances are displayed in Tables 1 and 2.

Furthermore, we set a time-window $[DT_j - 20, DT_j + 20]$ for each flight $j \in L$, and we randomly generated for each flight network two passenger scenarios. For each flight $j \in L$, the number of passengers was generated as follows. Given the average load factor LF, and the aircraft capacity AC, the number of passengers was arbitrarily drawn from $U[0.8LF \times AC, \min(AC, 1.2LF \times AC)]$. Hence, our testbed includes six large-scale instances QA_i^k, where the superscript $k = 1, 2$ refers to the passenger scenario.

The turn-times as well as the maintenance times are aircraft type-dependent. The corresponding values are displayed in Table 3. Furthermore, in alignment with the airline's current practice, we assumed that each aircraft should be offered an opportunity to undergo at least one maintenance check during the week.

6.3. Numerical results

Is this section, we assess the empirical performance of the proposed two-stage approach. All the our algorithms were coded in Microsoft Visual C++ (version 6.0), and the model (*MILP*) was solved using the CPLEX 12.6 solver. The computational experiments were carried out on an i7 dual core 1.8 GHz Personal Computer having 8 GB RAM.

6.3.1. Performance of Model MILP

In a first experiment, we assess the performance of Model *MILP*. Since the utility function depends on the parameter β, we empirically evaluated three different values of this parameter as follows. Given a connection (i,j) with $\bar{d}_i$ being the mean primary delay of flight i, the parameter β was set at three different values yielding a utility for $\bar{d}_i$ equal to 0.3, 0.5, and 0.7, respectively. Hence, we accordingly derived three utility functions that are denoted $u^{0.3}$, $u^{0.5}$, and $u^{0.7}$, respectively.

Since each instance Q_i $(i \in \{1, 2, 3\})$ includes 10 aircraft types, and Model *MILP* is defined for each aircraft type, we thus obtain a testbed of 90 instances. In this experiment, we set a CPU time limit of one hour. A summary of the results is displayed in Table 4, where we provide, for each instance, the optimal value $Z^*(u^q)$ and the required CPU time $T(u^q)$ (in s) for $q \in \{0.3, 0.5, 0.7\}$.

We see from Table 4 that Model *MILP* remarkably delivered optimal solutions for all instances, except for those instances that pertain to aircraft type *A*320-200. Not surprisingly, these unsolved instances are the largest ones and include 1056 legs and 35 aircraft. To solve these large-scale instances using Model *MILP*, we implemented the following *relaxation* approach.

6.3.1.1. A relaxation approach for large-scale instances. The procedure starts with relaxing the maintenance constraints (36) and (37) and optimally solves the relaxed model. Accordingly, we generate p aircraft rotations, where each rotation covers a subset of legs that should be served by a specific number of aircraft (see Proposition 1). Next, the set of p rotations is randomly partitioned into p' subsets in such a way that each subset is covered by 6–10 aircraft. Hence, each subset of rotations defines a reduced instance having a corresponding number of aircraft ranging between 6 and 10. Model *MILP* is used then for solving these instances in turn. The final solution is therefore obtained by considering the union of the generated aircraft maintenance feasible routes.

Table 5 displays the performance of the relaxation approach. We see that all the reduced instances were quickly solved to optimality.

6.3.2. Impact of the graph reduction

Using Remark 4, we dropped from the graph all connections whose corresponding aircraft idle time is larger than 24 h. Table 6 displays the impact on this reduction for a sample of three fleets A321, A332, and A333, and for three instances QA_1^1,

Table 1
Characteristics of the flight networks.

Network	Aircraft	Fleet type	Flights	Flight clusters	Stations
QA_1	141	10	2906	499	129
QA_2	151	10	3098	527	138
QA_3	164	10	3387	575	145

Table 2
Fleet data of the flight networks.

Fleet	Instance					
	QA_1		QA_2		QA_3	
	Aircraft	Flights	Aircraft	Flights	Aircraft	Flights
A320-200	35	1056	38	1112	38	1195
A321-200	8	267	7	214	8	250
A330-200	16	260	16	254	16	275
A330-300	13	239	11	238	13	265
A340-600	4	62	4	66	4	82
A350	3	36	7	96	9	98
A380-800	4	52	6	82	6	82
B777-200	9	124	9	113	9	130
B777-300	28	402	30	457	33	495
B787-8	21	408	24	466	28	515

Table 3
Turn-times and maintenance durations.

Fleet	Turn-time (min)	Maintenance duration (min)
A320-200	40	480
A321-200	40	480
A330-200	45	540
A330-300	45	540
A340-600	60	600
A350	60	600
A380-800	60	600
B777-200	45	540
B777-300	60	600
B787-8	45	540

Table 4a
Performance of Model *MILP* on QA_1-based instances.

Fleet	$u^{0.3}$		$u^{0.5}$		$u^{0.7}$	
	Z^*	Time (in s)	Z^*	Time (in s)	Z^*	Time (in s)
A320-200	[a]	3600	[a]	3600	[a]	3600
A321-200	163.44	5.93	206.52	6.78	235.46	7.44
A330-200	180.91	7.80	213.39	9.01	233.85	6.96
A330-300	166.54	15.84	200.42	26.18	222.46	5.68
A340-600	40.03	0.23	45.67	0.09	49.65	0.05
A350	27.85	0.05	31.54	0.04	33.79	0.07
A380-800	40.36	0.04	45.99	0.07	49.17	0.05
B777-200	87.49	0.39	104.20	0.32	113.55	0.38
B777-300	291.79	135.84	343.45	151.76	373.17	187.84
B787-8	295.48	36.18	346.15	34.66	375.84	35.24

[a] Unsolved after one hour CPU time.

QA_2^1 and QA_3^1, respectively. We see from this table that the reduction procedure enables a critical reduction of the CPU time while having a negligible impact on the objective function.

6.3.3. Robustness of the aircraft routes that are generated in Stage 1

We used *SimAir* to assess the robustness of the aircraft routes that are generated in Stage 1 upon solving the MIP models, and we compared them against the corresponding original solutions, which correspond to schedules that were actually

Table 4b
Performance of Model *MILP* model on QA_2-based instances.

Fleet	$u^{0.3}$		$u^{0.5}$		$u^{0.7}$	
	Z*	Time (in s)	Z*	Time (in s)	Z*	Time (in s)
A320-200	[a]	3600	[a]	3600	[a]	3600
A321-200	112.38	1.23	146.54	1.13	174.64	1.24
A330-200	174.60	137.21	205.93	33.60	228.58	96.32
A330-300	133.36	17.83	171.65	19.54	201.65	10.65
A340-600	40.69	0.13	47.85	0.09	55.43	0.12
A350	70.55	0.24	80.66	0.38	85.55	0.14
A380-800	56.33	0.27	67.11	0.23	74.28	0.21
B777-200	71.74	0.23	88.50	0.32	99.86	0.22
B777-300	301.73	259.55	367.35	207.21	408.93	456.44
B787-8	287.64	245.37	361.03	469.35	411.87	443.10

[a] Unsolved after one hour CPU time.

Table 4c
Performance of Model *MILP* on QA_3-based instances.

Fleet	$u^{0.3}$		$u^{0.5}$		$u^{0.7}$	
	Z*	Time (in s)	Z*	Time (in s)	Z*	Time (in s)
A320-200	[a]	3600	[a]	3600	[a]	3600
A321-200	132.48	3.18	173.70	3.42	206.64	1.16
A330-200	186.13	10.02	221.85	8.94	246.37	9.84
A330-300	164.25	8.05	198.72	8.02	228.03	22.86
A340-600	46.34	0.12	58.27	0.14	67.94	0.19
A350	70.94	0.41	82.36	0.62	89.69	0.57
A380-800	58.16	0.24	68.61	0.19	75.23	0.22
B777-200	79.74	0.97	99.20	1.09	113.21	0.83
B777-300	318.98	1757.27	389.37	1882.42	438.41	2188.28
B787-8	311.99	3499.78	386.70	3446.45	440.56	3025.60

[a] Unsolved after one hour CPU time.

Table 5
Performance of Model *MILP* on the set of reduced instances of fleet type *A*320-200.

Network	Sub Ins.	Flights	Aircraft	$u^{0.3}$		$u^{0.5}$		$u^{0.7}$	
				Z*	Time (in s)	Z*	Time (in s)	Z*	Time (in s)
QA_1	1	242	8	152.90	2.45	190.02	6.62	215.20	2.77
	2	244	8	154.39	1.94	190.92	1.17	216.53	1.35
	3	272	9	172.93	3.98	214.11	2.96	242.08	3.28
	4	302	10	192.17	6.09	238.04	3.98	269.01	4.97
QA_2	1	276	10	155.37	8.08	197.46	6.65	230.43	3.37
	2	303	10	172.26	5.83	219.24	4.35	255.60	5.63
	3	278	9	150.38	5.60	194.69	4.18	230.09	4.72
	4	256	9	141.31	2.23	180.94	3.98	212.75	5.10
QA_3	1	210	7	113.43	2.92	147.93	2.93	175.00	2.18
	2	268	8	140.88	11.53	181.70	83.16	217.78	25.68
	3	314	10	136.38	29.83	217.83	45.20	257.78	32.72
	4	190	6	99.56	2.23	131.94	1.48	156.90	0.96
	5	213	7	109.89	4.60	142.42	4.28	174.72	3.76

implemented by the airline company. In Table 7, we provide a summary of the main results, and the average performance measures PM_1, PM_2 and PM_3, respectively. We also display for each instance QA_i^k the percentage deviation $100 \times \frac{PM_i^k(S1)-PM_i^k(original)}{P_i^k(S1)}$, $i = 1\ldots3$, $k = 1, 2$, where $PM_i^k(S1)$ and $PM_i^k(original)$ refer to the performance measure of the solution that is generated in Stage 1 and the original one, respectively.

Examining the results in this table, we see that the new aircraft routes are more robust than the original ones for all instances. The improvement is very significant for both instances QA_1^1 and QA_1^2. By contrast, a marginal improvement is achieved for the remaining instances. Interestingly, we observe that variations in the parameter β enables different solutions that exhibit slightly different performances.

Table 6a
Impact of graph reduction on QA_1-based instances.

		A321-200	A330-200	A330-300
Number of arcs	Before reduction	17,255	13,649	14,220
	After reduction	2935	2327	2411
	Reduction percentage	82.99%	82.95%	83.05%
Objective function	Before reduction	163.44	180.91	165.54
	After reduction	163.10	180.91	166.35
CPU time	Before reduction	181.01	218.33	221.72
	After reduction	5.93	7.91	15.84
	Reduction percentage	96.72%	96.38%	92.86%

Table 6b
Impact of graph reduction on QA_2-based instances.

		A321-200	A330-200	A330-300
Number of arcs	Before reduction	11,049	13,698	13,213
	After reduction	1855	2389	2243
	Reduction percentage	83.21%	82.56%	83.02%
Objective function	Before reduction	112.39	174.80	135.74
	After reduction	112.38	174.60	133.36
CPU time	Before reduction	12.99	407.56	205.55
	After reduction	1.23	137.21	17.83
	Reduction percentage	90.53%	66.33%	91.33%

Table 6c
Impact of graph reduction on QA_3-based instances.

		A321-200	A330-200	A330-300
Number of arcs	Before reduction	15,013	16,505	15,692
	After reduction	2580	2831	2687
	Reduction percentage	82.81%	82.85%	82.88%
Objective function	Before reduction	132.56	188.29	164.68
	After reduction	132.48	186.13	164.25
CPU time	Before reduction	38.18	148.42	692.90
	After reduction	3.18	10.02	8.05
	Reduction percentage	91.67%	93.25%	98.84%

Table 7a
Performance of stage 1 on QA_1-based instances.

		PM_1 (%)	Deviation (%)	PM_2 (min)	Deviation (%)	PM_3	Deviation (%)
QA_1^1	*Original*	75.95	*	17551.67	*	10096.07	*
	$u^{0.3}$	79.92	5.2	14551.89	17.1	9208.46	8.8
	$u^{0.5}$	79.89	5.2	14567.72	17.0	9348.52	7.4
	$u^{0.7}$	79.89	5.2	14541.32	17.2	9337.84	7.5
QA_1^2	*Original*	76.07	*	17511.67	*	10535.8	*
	$u^{0.3}$	79.35	4.3	15138.74	13.6	7886.04	25.2
	$u^{0.5}$	79.35	5.0	14594.66	16.7	7724.48	26.7
	$u^{0.7}$	79.89	5.0	14541.32	17.0	7669.04	27.2

6.3.4. Performance of the simulation-based retiming approach

For each instance, we selected the solution that was derived in Stage 1 and that exhibits the highest on-time performance (we use the total delays for breaking ties), and we used *SimAir* to assess the robustness of the solutions generated after invoking the retiming procedure. In Table 8, we provide for each instance QA_i^k the percentage deviation $100 \times \frac{PM_i^k(OS)-PM_i^k(original)}{P_i^k(OS)}$, $i = 1, \ldots, 3$, $k = 1, 2$, where *OS* refers to the optimization-simulation approach.

We see from Table 8 that the proposed solutions exhibit high solution quality. Indeed, the on-time performance was improved by 9.8–16.0%, the cumulative delay was reduced by 25.4–33.1%, and the number of delayed passengers was reduced by 8.2–51.6%.

Table 7b
Performance of stage 1 on QA_2-based instances.

		PM_1 (%)	Deviation (%)	PM_2 (min)	Deviation (%)	PM_3	Deviation (%)
QA_2^1	*Original*	68.39	*	30104.40	*	13588.58	*
	$u^{0.3}$	68.91	0.8	29555.32	1.8	13163.24	3.1
	$u^{0.5}$	68.94	0.8	29430.14	2.2	13124.10	3.4
	$u^{0.7}$	68.89	0.7	29506.00	2.0	13084.10	3.7
QA_2^2	*Original*	68.33	*	30213.66	*	14640.86	*
	$u^{0.3}$	68.93	0.9	29458.36	2.5	13866.30	5.3
	$u^{0.5}$	68.96	0.9	29446.49	2.5	13998.26	4.4
	$u^{0.7}$	68.91	0.8	29520.04	2.3	13931.72	4.8

Table 7c
Performance of stage 1 on QA_3-based instances.

		PM_1 (%)	Deviation (%)	PM_2 (min)	Deviation (%)	PM_3	Deviation (%)
QA_3^1	*Original*	65.17	*	33471.17	*	15170.40	*
	$u^{0.3}$	65.67	0.8	32801.40	2.0	15079.72	0.6
	$u^{0.5}$	65.65	0.7	32864.94	1.8	15120.74	0.3
	$u^{0.7}$	65.64	0.7	32938.42	1.6	15059.98	0.7
QA_3^2	*Original*	65.14	*	33567.30	*	15964.13	*
	$u^{0.3}$	65.70	0.8	32768.75	2.4	15410.80	3.5
	$u^{0.5}$	65.72	0.9	32749.02	2.4	15652.90	1.9
	$u^{0.7}$	65.65	0.8	32926.57	1.9	15832.84	0.8

Table 8
Performance of the simulation-based retiming approach.

		PM_1 (%)	Deviation (%)	PM_2 (min)	Deviation (%)	PM_3	Deviation (%)	CPU (s)
QA_1^1	*Original*	75.95	*	17551.67	*	10096.07	*	754.02
	$u^{0.7}$	83.44	9.9	11829.90	32.6	9268.60	8.2	
QA_1^2	*Original*	76.07	*	17511.67	*	10535.80	*	
	$u^{0.7}$	83.50	9.8	11789.64	32.7	5094.68	51.6	
QA_2^1	*Original*	68.39	*	30104.40	*	13588.58	*	2289.12
	$u^{0.5}$	75.36	10.2	22458.74	25.4	8945.26	34.2	
QA_2^2	*Original*	68.33	*	30213.66	*	14640.86	*	
	$u^{0.5}$	8854.98	10.3	22458.74	25.7	8854.98	39.5	
QA_3^1	*Original*	65.17	*	33471.17	*	15170.40	*	1954.04
	$u^{0.3}$	75.55	15.9	22443.60	32.9	9,890.66	34.8	
QA_3^2	*Original*	65.14	*	33567.30	*	15964.13	*	
	$u^{0.3}$	75.55	16.0	22443.60	33.1	9706.26	39.2	

Table 9
Average gap of the departures times.

Network	Average difference (min)
QA_1	13.63
QA_2	15.92
QA_3	14.71

As a final experiment, we analyzed the amount by which the schedule was altered after running the retiming procedure. Toward this end, we computed for each flight a gap that represents the absolute value of the difference between the original departure time and the revised one. The results are displayed in Table 9.

We can see from this table, that while most departure times were changed, the gaps are relatively small (around 15 min). It is remarkable to observe that although these changes are modest, they greatly impacted the schedule's robustness.

7. Conclusion

In this paper, we have addressed the robust weekly aircraft maintenance routing and retiming problem. Despite its crucial importance, this tactical planning problem has received scant attention in the literature. To solve this large-scale, stochastic, combinatorial optimization problem, we proposed a hybrid optimization-simulation approach. The proposed approach decomposes the problem into two subproblems that are consecutively solved. First, maintenance feasible aircraft routes are derived upon solving a novel compact mixed-integer programming model. Next, flight departure times are iteratively adjusted using a Monte-Carlo simulation procedure with the aim of enhancing the solution's robustness, while accommodating synchronization and maintenance constraints. We presented the results of extensive computational experiments on instances having up to 3387 flights and 164 aircraft, and we demonstrated that the proposed MIP model enables the construction of judicious maintenance feasible aircraft routes while requiring reasonable CPU times. For the largest fleets, we implemented a relaxation heuristic and used it to partition the aircraft routing problem into smaller problems where each problem is solved separately using the proposed MIP model. We used *SimAir* as a simulation tool to assess the robustness of the generated schedules. We found that for the solutions produced by our approach in comparison to the original solutions implemented by the airline, the on-time performance was improved by 9.8–16.0%, the cumulative delay was reduced by 25.4–33.1%, and the number of delayed passengers was reduced by 8.2–51.6%.

As a future research direction, we recommend conducting a comprehensive theoretical and computational study of surrogate robust measures for aircraft routing. Whereas such surrogate robustness measures were extensively investigated in the project scheduling literature (see Chtourou and Haouari, 2008; Hazır et al., 2010), their application in robust aircraft routing is almost void. It is expected that the development of more efficient slack utility functions will enable generating more robust aircraft routes via Stage 1, and thereby enhance the overall efficacy of the proposed simulation-optimization procedure.

A second future research direction is to integrate flight retiming, aircraft maintenance routing, and crew pairing into a single model in order to simultaneously derive robust aircraft routes and crew pairings. The integration of these processes will enable airlines to effectively enhance their on-time performance through a coordinated planning of their most vital assets (aircraft and crews).

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